

Senthamizhvelan M

+91 8015404762 | senthamizhvelan13@gmail.com | github.com/senthamizhvelan04

PROFESSIONAL SUMMARY

AI & Data Science graduate specializing in agentic AI systems, RAG architectures, and LLM tool-orchestration. Built and shipped a multi-provider autonomous research agent and a production RAG pipeline reducing information retrieval time by 70%. Proficient in Python, SQL, and FastAPI; comfortable owning a system from architecture through deployment.

EXPERIENCE

CITPL (Cavin Info Tech Private Limited) <i>AI/ML Intern</i>	Feb 2026 – Aug 2026 <i>6 Months</i>
- Automated internal business workflows using n8n, cutting manual processing time by 20% - Built and deployed RAG-based systems for internal knowledge retrieval, integrating + REST API endpoints - Implemented Model Context Protocol (MCP) integrations to connect LLM workflows with internal tools and data sources	

TECHNICAL SKILLS

Programming: Python, SQL, Java
Machine Learning: Pandas, NumPy, NLP, RAG (Retrieval-Augmented Generation), Agentic AI / Tool-Calling Orchestration
Web Technologies: Flask, FastAPI, REST API, HTML, CSS
Databases: MySQL, PostgreSQL
Tools: Git, GitHub, MCP (Model Context Protocol), n8n, Data Structures & Algorithms

PROJECTS

Autonomous AI Research Agent <i>Python, Tool Calling</i> GitHub
- Architected a fully autonomous multi-step research agent implementing a Plan → Act → Observe → Reflect reasoning loop with native LLM function/tool calling - Integrated Groq (Llama 3.3 70B) as the primary reasoning engine with automatic fallback to Google Gemini, cutting task failure rate to near-zero under free-tier API rate limits via retry-with-backoff logic - Orchestrated multiple tools (web search, page fetching, calculation) and built a source-verification layer that cross-checks facts, flags conflicting information, and requires citations to reduce hallucination - Designed bounded execution (iteration caps, per-question retry limits) so the agent produces a structured, cited Markdown report and gracefully reports incomplete work instead of failing silently
Neural RAG System – AI Internal Knowledge Assistant <i>Python, FastAPI, Vector DB</i> GitHub
- Developed an intelligent knowledge management system using RAG architecture for instant company information access across 500 + internal documents - Implemented FastAPI backend with vector database integration and semantic search, achieving sub 2 second query response times - Designed RESTful APIs with authentication, reducing information retrieval time by 70%

EDUCATION

CK College of Engineering and Technology, Cuddalore <i>Bachelor of Technology in Artificial Intelligence & Data Science</i>	2026 <i>CGPA: 8.5/10</i>
Anandhan Memorial Metric HSS <i>Higher Secondary (12th)</i>	<i>87.53%</i>
Anandhan Memorial Metric HSS <i>Secondary (10th)</i>	<i>84.6%</i>